

IBMP

Pain Medicine

Online examination / Question paper / Set A

FACULTY-REVIEW DRAFT - approve before examination release

Section	Questions	Marks each	Total
A. MCQs	40	1	40
B. Image-based	10	3	30
C. Short answers	10	1	10
TOTAL	60	-	80

Viva and project are assessed separately and are not included in these 80 marks.

Candidate name: _____ Student ID: _____

Exam date: _____ Suggested duration: 120 minutes

Candidate instructions

- Attempt all 60 questions. Section A requires one best answer for each MCQ.
- Section B contains ten original figures. Answer all three parts of each image question; each part carries 1 mark unless a split is stated in the key.
- Section C requires a brief answer: usually one term, one example or one sentence.
- Proposed scoring: correct MCQ = 1; incorrect/unanswered MCQ = 0. No negative marking is proposed. IBMP must confirm the final rules and duration before release.
- All cases and plotted observations are fictional. Images are teaching schematics, not patient photographs or diagnostic scans. Interpret each figure together with its stem.
- This written assessment tests knowledge and clinical reasoning. It does not independently certify hands-on procedural competence.

Reference rationales and the confidential marking guide are supplied separately to faculty.

Section A | Multiple-choice questions

Q01-Q05 | Choose one best answer | 1 mark each

Q01 After a pinprick, a patient perceives an immediate, well-localised sharp pain before a slower burning sensation. Which fibres primarily carry the first sensation? **[1 mark]**

- g) Preganglionic autonomic B fibres
- h) Unmyelinated C fibres
- i) Thinly myelinated A-delta fibres
- j) Large myelinated A-alpha motor fibres

Q02 Repeated nociceptive input increases the responsiveness of neurons within the dorsal horn, including responses to previously subthreshold input. What process does this describe? **[1 mark]**

- k) Central sensitisation
- l) Wallerian degeneration
- m) Peripheral nerve conduction block
- n) Descending inhibition

Q03 Light clothing brushing a painful area produces pain, although the same contact is comfortable elsewhere. What is this finding called? **[1 mark]**

- o) Hyperalgesia
- p) Paresthesia
- q) Hypoesthesia
- r) Allodynia

Q04 A clinician records a patient's pain intensity as 7/10. Which additional assessment is most useful for judging the wider impact of persistent pain? **[1 mark]**

- s) The number of analgesic brands previously prescribed
- t) The number of abnormalities in an imaging report alone
- u) The patient's height and dominant hand alone
- v) Interference with sleep, activity and participation

Q05 A patient with chronic primary pain avoids all movement because they believe any discomfort means new injury. Which approach best addresses this barrier? **[1 mark]**

- w) Recommend indefinite rest while repeating unchanged investigations
- x) Explore the belief and agree a paced, graded return to meaningful activity
- y) Delay rehabilitation until every pain sensation has disappeared
- z) Confirm that all pain during movement represents tissue injury

Section A | Multiple-choice questions

Q06-Q10 | Choose one best answer | 1 mark each

Q06 An adult with decision-making capacity declines an elective pain procedure after discussing its benefits, risks and alternatives. What is the most appropriate response? **[1 mark]**

- aa) Respect the decision, document it and discuss acceptable alternatives
- bb) Withhold all further pain care until the patient agrees
- cc) Treat the refusal as evidence of impaired capacity
- dd) Proceed if a relative signs the consent form instead

Q07 After explaining a new analgesic plan, which action best checks whether the clinician has communicated clearly? **[1 mark]**

- ee) Assume understanding if the patient has previously used pain medicines
- ff) Ask the patient to explain the plan and key precautions in their own words
- gg) Ask the patient to memorise the medicine leaflet verbatim
- hh) Ask only whether the patient understands the prescription

Q08 A patient with chronic kidney disease and recent dehydration requests oral ibuprofen. Which adverse effect is of particular concern? **[1 mark]**

- ii) Further renal impairment from reduced renal prostaglandin activity
- jj) Permanent suppression of pancreatic insulin release
- kk) Direct irreversible blockade of skeletal neuromuscular transmission
- ll) Acute opioid withdrawal from mu-receptor antagonism

Q09 A patient takes paracetamol tablets and an over-the-counter cold preparation that also contains paracetamol. What is the key medication-safety concern? **[1 mark]**

- mm) The second preparation reliably protects against liver toxicity
- nn) Unrecognised duplication can produce a hepatotoxic total dose
- oo) Combination products prevent paracetamol absorption completely
- pp) Only prescription paracetamol contributes to the daily total

Q10 Before initiating gabapentin in an older patient with neuropathic pain, which factor is especially important when selecting the dose and dosing interval? **[1 mark]**

- qq) Hepatic enzyme induction alone
- rr) Serum albumin alone
- ss) Renal function
- tt) Body weight without renal assessment

Section A | Multiple-choice questions

Q11-Q15 | Choose one best answer | 1 mark each

Q11 An adult has painful diabetic peripheral neuropathy and no contraindication to the options listed. Which medicine is an established initial treatment option? **[1 mark]**

- uu) Codeine
- vv) Prednisolone
- ww) Diazepam
- xx) Duloxetine

Q12 A patient receiving an opioid is also taking a benzodiazepine. Which combined adverse effect requires particular attention? **[1 mark]**

- yy) Loss of all opioid analgesic effect through receptor blockade
- zz) Sedation and respiratory depression
- aaa) Reliable reversal of opioid-related constipation
- bbb) Reduced risk of falls through improved coordination

Q13 A stable patient receiving long-term opioids for chronic non-cancer pain has no immediate life-threatening toxicity. A review finds that risks now outweigh benefits. What is the preferred approach? **[1 mark]**

- ccc) Agree an individualised, supported reduction plan and optimise other care
- ddd) Increase the dose automatically to prevent any future discomfort
- eee) Stop treatment abruptly without discussing withdrawal or follow-up
- fff) Discharge the patient solely because a taper may be needed

Q14 Before starting an opioid for chronic non-cancer pain, which plan best supports responsible prescribing? **[1 mark]**

- ggg) Use a signed agreement instead of clinical assessment and monitoring
- hhh) Promise indefinite continuation regardless of function or harms
- iii) Agree measurable goals, assess risks, explain precautions and arrange review
- jjj) Use pain intensity as the only follow-up measure and omit adverse-effect review

Q15 Which presentation is the best match for considering an epidural injection of local anaesthetic and steroid under NICE NG59? **[1 mark]**

- kkk) Neurogenic claudication due to central canal stenosis
- lll) Acute, severe sciatica after clinical assessment
- mmm) Chronic non-specific axial back pain without sciatica
- nnn) Asymptomatic lumbar degenerative findings on MRI

Section A | Multiple-choice questions

Q16-Q20 | Choose one best answer | 1 mark each

Q16 A patient is being assessed for radiofrequency treatment of suspected lumbar facet-mediated pain. Which finding most directly supports proceeding with a medial branch pathway? **[1 mark]**

- ooo) A positive screening questionnaire without examination
- ppp) Facet degeneration on imaging without a matching clinical assessment
- qqq) Relief after an unrelated peripheral joint injection
- rrr) A clinically meaningful response to a properly performed diagnostic medial branch block

Q17 Conventional thermal radiofrequency treatment for lumbar facet-mediated pain is intended primarily to affect which structure? **[1 mark]**

- sss) Medial branches carrying sensation from the facet joints
- ttt) The spinal cord itself
- uuu) The main femoral nerve within the pelvis
- vvv) The nucleus pulposus of every lumbar disc

Q18 A patient asks for an intradiscal procedure because an MRI reports disc degeneration. What is the most appropriate next step? **[1 mark]**

- www) Promise restoration of normal disc anatomy after any intradiscal treatment
- xxx) Select a procedure solely according to the darkest disc on the scan
- yyy) Offer the procedure because degeneration proves the disc is the pain generator
- zzz) Correlate symptoms and examination, review alternatives and explain procedure-specific evidence

Q19 Which statement is most accurate when counselling a patient about image-guided epidural steroid injection? **[1 mark]**

- aaaa) Fluoroscopy makes vascular or neurological injury impossible
- bbbb) Image guidance assists targeting but does not eliminate serious complications
- cccc) A normal prior MRI removes the need for procedure-specific consent
- dddd) Image guidance eliminates the need to check medicines and infection risk

Q20 A patient develops rapidly progressive leg weakness and new urinary retention after a spinal pain intervention. What is the most appropriate response? **[1 mark]**

- eeee) Advise increasing sedating medicines and waiting at home
- ffff) Schedule routine review after the next course of physiotherapy
- gggg) Arrange immediate emergency assessment for a possible compressive neurological complication
- hhhh) Reassure the patient that this is an expected long-term effect

Section A | Multiple-choice questions

Q21-Q25 | Choose one best answer | 1 mark each

Q21 Local anaesthetics produce peripheral nerve blockade mainly through which mechanism? **[1 mark]**

- iiii) Inhibition of voltage-gated sodium channels
- jjjj) Destruction of acetylcholine receptors at every neuromuscular junction
- kkkk) Inhibition of cyclo-oxygenase in the liver
- llll) Activation of mu-opioid receptors

Q22 A patient attending for an elective knee steroid injection has a hot swollen knee and fever. What is the best immediate decision? **[1 mark]**

- mmmm) Inject steroid first and investigate only if the swelling remains
- nnnn) Give a larger steroid dose because fever indicates severe inflammation
- oooo) Defer the steroid injection and urgently assess for joint infection
- pppp) Proceed if the patient signs an additional consent form

Q23 Which intervention may be considered as an adjunct for selected patients with refractory upper abdominal visceral pain from pancreatic cancer? **[1 mark]**

- qqqq) Occipital nerve block
- rrrr) Celiac plexus block or neurolysis
- ssss) Suprascapular nerve block
- tttt) Lumbar facet medial branch block

Q24 Which examination finding most strongly supports an active myofascial trigger point? **[1 mark]**

- uuuu) A tender spot within a taut muscle band that reproduces the patient's familiar pain
- vvvv) Painless joint crepitus with preserved movement
- wwww) An isolated absent ankle reflex without local muscle tenderness
- xxxx) A dermatomal sensory deficit without a palpable muscle finding

Q25 During an ultrasound-guided peripheral intervention, which practice best supports safety? **[1 mark]**

- yyyy) Treat ultrasound guidance as a replacement for monitoring and asepsis
- zzzz) Advance whenever the needle shaft is visible even if the tip is not located
- aaaaa) Identify relevant vessels and maintain awareness of the needle tip throughout advancement
- bbbbb) Assume that a dark circular structure cannot be vascular

Section A | Multiple-choice questions

Q26-Q30 | Choose one best answer | 1 mark each

Q26 A patient with advanced cancer reports pain worsened by fear, financial worries and isolation. Which assessment best reflects a comprehensive palliative approach? **[1 mark]**

- ccccc) Physical symptoms together with psychological, social and spiritual concerns
- ddddd) The latest pain score without contextual questions
- eeee) Analgesic dose alone as a substitute for patient assessment
- fffff) Tumour measurements alone

Q27 A patient presents with severe cancer pain. Which interpretation of the WHO analgesic approach is most appropriate? **[1 mark]**

- ggggg) Use exactly the same medicine and schedule for every patient
- hhhhh) Require a prolonged trial of paracetamol alone before considering stronger analgesia
- iiiii) Select analgesia appropriate to severity, including an opioid when indicated, with reassessment
- jjjjj) Avoid opioids in all cancer pain because dependence is possible

Q28 Which patient is the most appropriate candidate for specialist assessment of intrathecal analgesic therapy? **[1 mark]**

- kkkkk) A patient seeking a guarantee that an implanted device has no complications
- lllll) Any patient with a single abnormal blood test but no pain
- mmmmm) A patient with mild pain controlled by simple treatment who wants no follow-up
- nnnnn) A patient with refractory cancer pain or intolerable systemic adverse effects despite optimised care

Q29 For a dying patient with pain, what is the most appropriate guiding principle for analgesic adjustment? **[1 mark]**

- ooooo) Escalation solely to achieve unconsciousness regardless of symptoms or goals
- ppppp) Withholding all analgesia because respiratory adverse effects are possible
- qqqqq) Continuation of an ineffective regimen to avoid any discussion with the family
- rrrrr) Proportionate symptom relief consistent with the patient's goals, with appropriate reassessment

Q30 When may specialist palliative care appropriately be integrated into cancer treatment? **[1 mark]**

- sssss) Only after all anticancer therapy has stopped
- ttttt) Alongside active cancer treatment when symptom or support needs warrant it
- uuuuu) Only during the final 24 hours of life
- vvvvv) Only when the patient no longer wishes to speak with the oncology team

Section A | Multiple-choice questions

Q31-Q35 | Choose one best answer | 1 mark each

Q31 Which new symptom in a patient with low back pain most strongly requires urgent evaluation for cauda equina syndrome? **[1 mark]**

- wwwww) Pain that improves with a comfortable position
- xxxxx) Intermittent muscle ache after unfamiliar exercise
- yyyyy) Urinary retention with saddle sensory change
- zzzzz) Mild stiffness after prolonged sitting

Q32 Persistent burning pain with sensory change in a healed herpes-zoster distribution is best classified as which type of pain? **[1 mark]**

- aaaaa) Pain from an acutely inflamed joint
- bbbbb) Neuropathic pain
- ccccc) Purely visceral nociceptive pain
- ddddd) Pain defined solely by psychosocial factors

Q33 Which statement about diagnosing complex regional pain syndrome is most accurate? **[1 mark]**

- eeeeee) A normal radiograph excludes the syndrome
- fffff) Use clinical criteria incorporating symptoms, examination signs and exclusion of a better explanation
- ggggg) Pain after any fracture is sufficient without other features
- hhhhh) A single positive bone scan establishes the diagnosis in every patient

Q34 A patient has widespread persistent pain, fatigue and unrefreshing sleep, with no identified lesion or active tissue disease adequately explaining the symptoms. Which pain mechanism is most consistent with a fibromyalgia presentation? **[1 mark]**

- iiiiii) Pain caused only by acute tissue inflammation
- jjjjjj) Pain caused only by a single transected peripheral nerve
- kkkkk) Pain that cannot exist unless imaging is abnormal
- lllll) Nociceptive pain

Q35 Which outcome best represents a useful rehabilitation goal for a patient with chronic primary pain? **[1 mark]**

- mmmmm) Progressively increasing an agreed meaningful activity while monitoring tolerability
- nnnnn) Eliminating all movement that produces any temporary discomfort
- ooooo) Increasing treatment visits regardless of function
- ppppp) Complete bed rest until the pain score has remained zero for a month

Section A | Multiple-choice questions

Q36-Q40 | Choose one best answer | 1 mark each

Q36 An adult has a clinical diagnosis of trigeminal neuralgia and no contraindication to the medicines listed. Which drug is recommended as initial treatment in NICE CG173? **[1 mark]**

- qqqqqq) Morphine
- rrrrrr) Pregabalin
- ssssss) Duloxetine
- tttttt) Carbamazepine

Q37 Before permanent spinal cord stimulation for an appropriate chronic neuropathic pain indication, which pathway is most suitable? **[1 mark]**

- uuuuuu) Selection solely by the number of prior prescriptions
- vvvvvv) Implantation based only on pain duration without assessing expectations
- wwwwww) Multidisciplinary selection, psychosocial assessment and an appropriate stimulation trial
- xxxxxx) Implantation without discussing maintenance or possible device complications

Q38 A patient with an intrathecal opioid pump reports a device alarm and sudden loss of analgesia. What is the most appropriate advice? **[1 mark]**

- yyyyyy) Seek urgent assessment by the responsible specialist or emergency service
- zzzzzz) Wait until the next routine refill regardless of symptoms
- aaaaaaa) Assume the alarm is unrelated because the wound looks normal
- bbbbbbb) Try to refill or reprogramme the pump at home

Q39 A clinic advertises an injected cell-based product as a guaranteed cure for degenerative back pain. What is the most appropriate professional response? **[1 mark]**

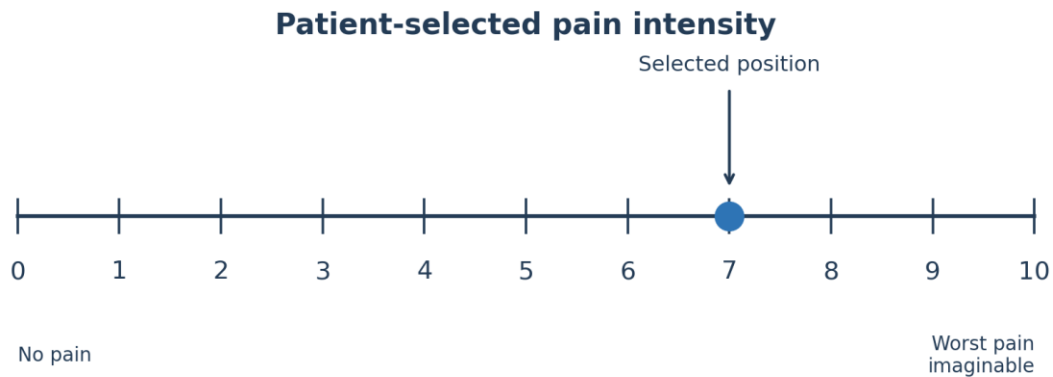
- ccccccc) Accept the claim if the clinic displays patient testimonials
- ddddddd) Treat a trial-registry listing as proof of efficacy and marketing approval
- eeeeeee) Evaluate product-specific evidence, regulatory status, risks and uncertainty before counselling
- ffffff) Assume that autologous products are risk-free and exempt from oversight

Q40 Which evidence would most strengthen confidence that a new pain therapy offers a clinically useful benefit? **[1 mark]**

- ggggggg) A laboratory mechanism without patient outcome data
- hhhhhhh) A clinic's claim that its method cannot be evaluated by comparison
- iiiiiii) A promotional video showing only selected satisfied patients
- jjjjjjj) Independent, well-conducted comparative trials reporting function, pain, harms and durability

Section B | Image-based questions

Q41 A patient selects the marked position on the scale below during an assessment of persistent pain. [3 marks]



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 1 | Original schematic; not to scale.

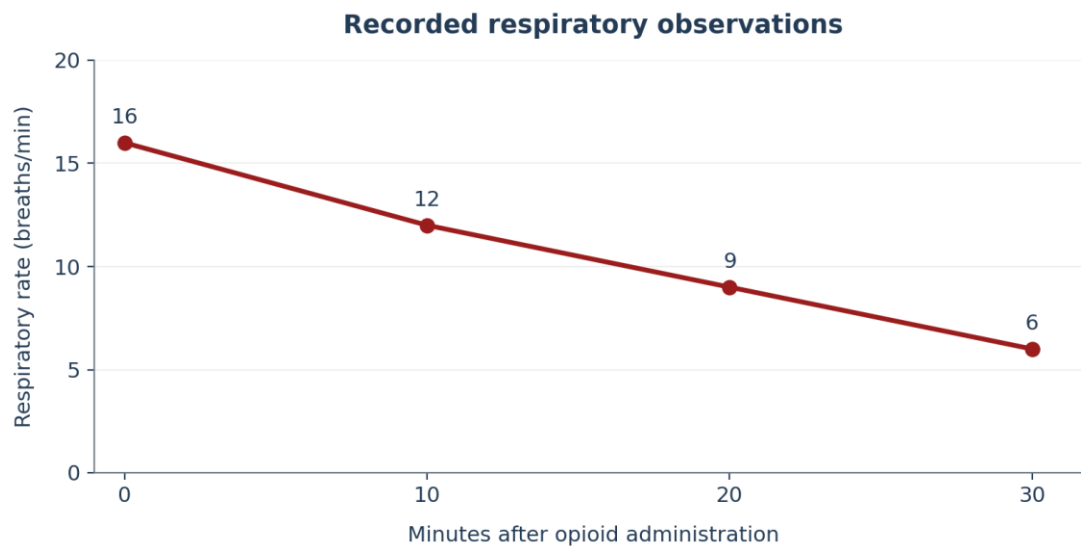
kkkkkkk) Record the patient's pain intensity score. [1 mark]

lllllll) Name one important dimension of pain impact not measured by this scale alone. [1 mark]

mmmmmmm) Write one focused question that would assess the dimension you named. [1 mark]

Section B | Image-based questions

Q42 An adult becomes increasingly difficult to rouse after an opioid dose. No other sedative has been given. The respiratory observations are shown below. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 2 | Original schematic; not to scale.

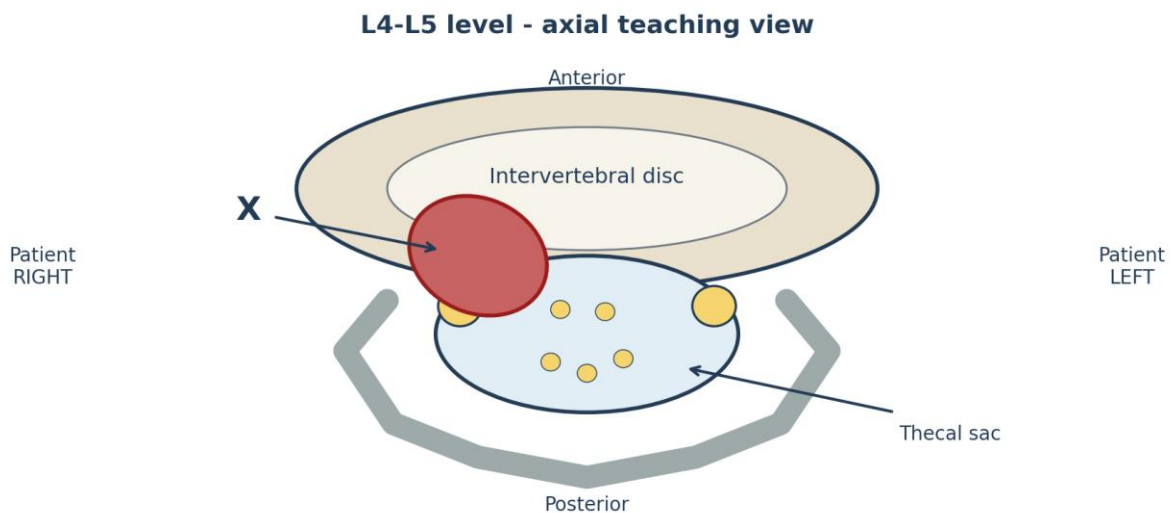
nnnnnnn) Identify the most concerning opioid-related complication. [1 mark]

oooooooo) State the immediate clinical priority; do not give a drug dose. [1 mark]

ppppppp) Name the opioid antagonist that may be required according to the emergency protocol.
[1 mark]

Section B | Image-based questions

Q43 An adult has right-sided radiating leg pain. The original axial teaching schematic represents the L4-L5 disc level; X marks an abnormality contacting the right traversing root. Patient right is shown on the left of the image. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 3 | Original schematic; not to scale.

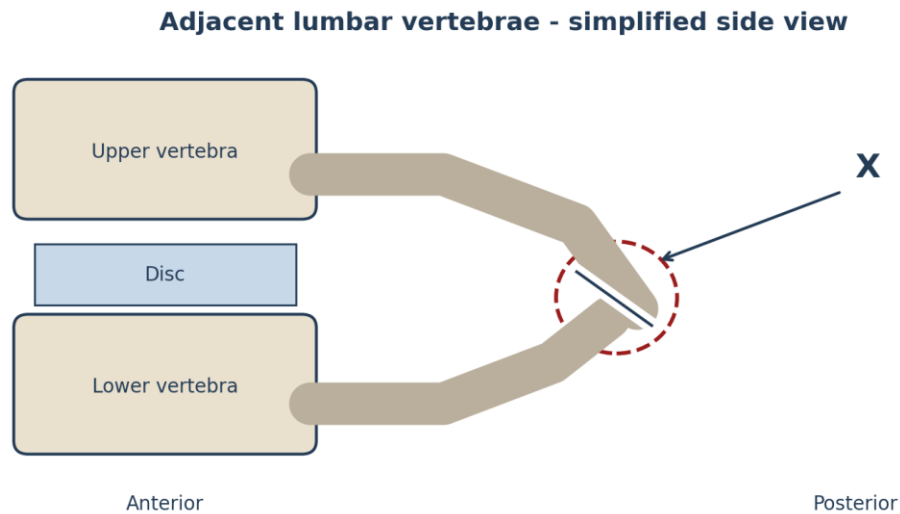
qqqqqqq) Identify the abnormality marked X. [1 mark]

rrrrrrr) Name the nerve root most likely affected at this location. [1 mark]

sssssss) State one new bladder or saddle-region symptom that would require urgent assessment. [1 mark]

Section B | Image-based questions

Q44 This simplified side-view schematic shows a posterior articulation between adjacent lumbar vertebrae. The articulation is marked X. [3 marks]



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 4 | Original schematic; not to scale.

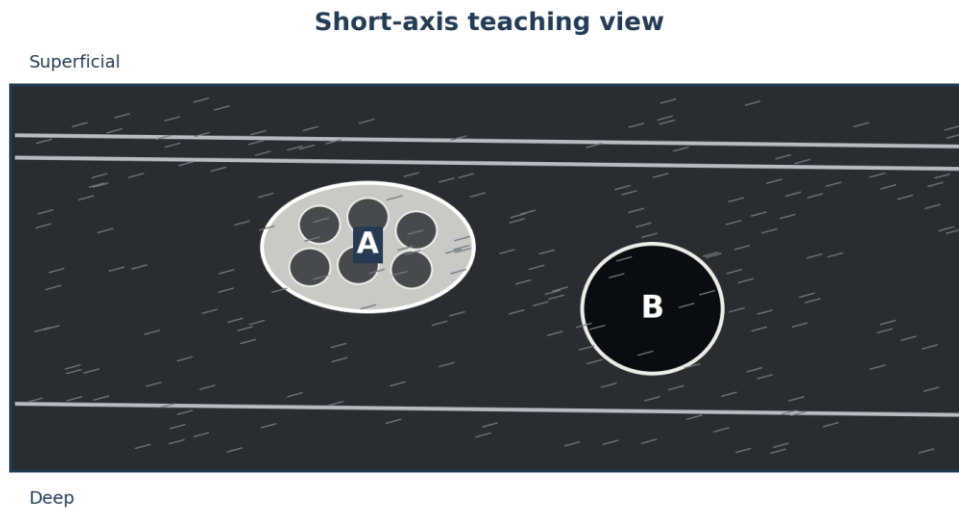
tttttt) Name the articulation marked X. [1 mark]

uuuuuuu) Name the nerve branches that provide its relevant sensory innervation. [1 mark]

vvvvvvv) Name the diagnostic/prognostic block commonly used before considering radiofrequency treatment of those branches. [1 mark]

Section B | Image-based questions

Q45 The teaching schematic below depicts a superficial peripheral nerve and a nearby pulsatile vessel in short axis. Structure A has multiple small internal fascicles; structure B is pulsatile. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 5 | Original schematic; not to scale.

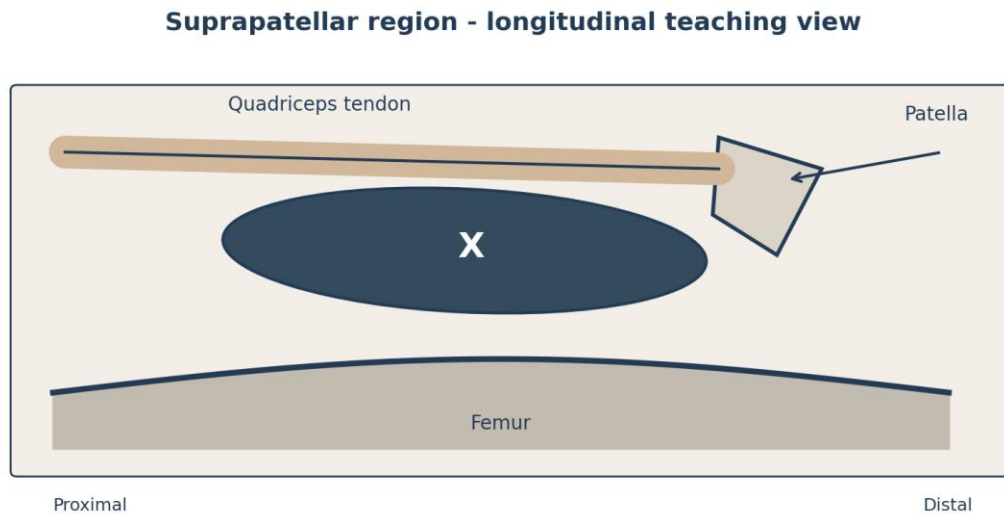
wwwwwww) Identify structure A. [1 mark]

xxxxxxx) Identify structure B and name one ultrasound mode that can help confirm blood flow.
[1 mark]

yyyyyyy) State one needle-visualisation safety principle for an ultrasound-guided intervention. [1
mark]

Section B | Image-based questions

Q46 A patient has a hot swollen knee and fever. This schematic longitudinal view shows the suprapatellar region; X marks the fluid-containing recess. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 6 | Original schematic; not to scale.

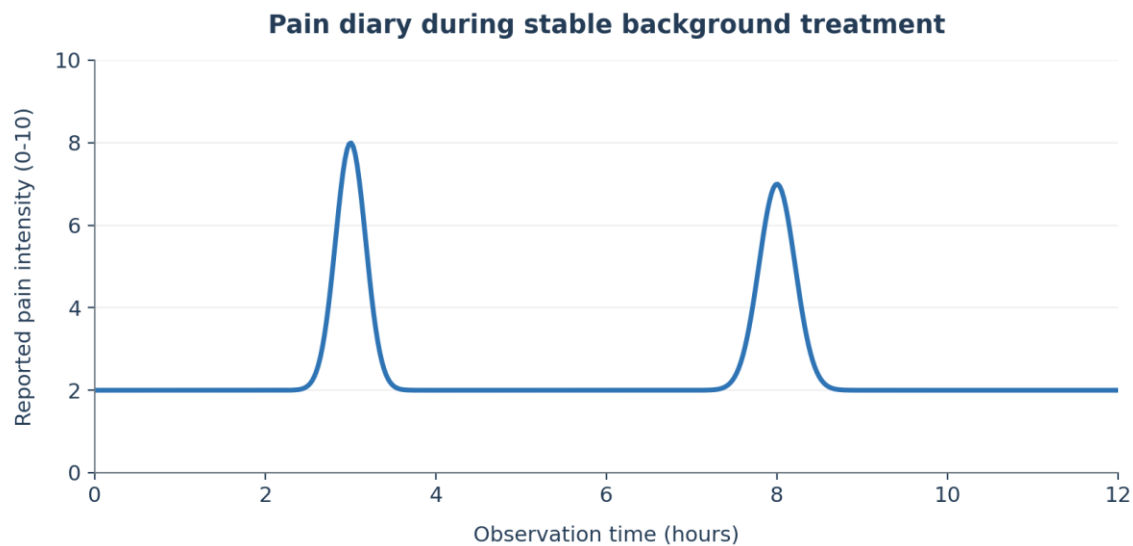
zzzzzz) Name the finding represented by the distended recess at X. [1 mark]

aaaaaaa) Name the serious diagnosis that must be considered before an elective steroid injection. [1 mark]

bbbbbbbb) State the appropriate next investigation/action to address that concern. [1 mark]

Section B | Image-based questions

Q47 A patient with cancer reports adequate background pain control on a stable regimen. The chart shows brief episodes of severe pain with return to the usual background level. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 7 | Original schematic; not to scale.

ccccccc) Name the pain pattern illustrated by the peaks. [1 mark]

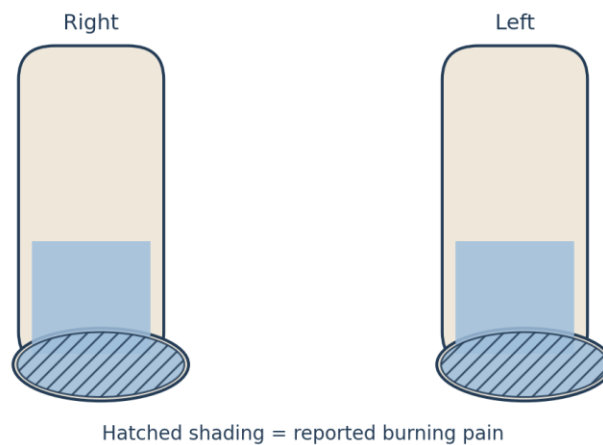
ddddddd) State the general medication strategy for an episode; do not specify a dose. [1 mark]

eeeeeee) If episodes become frequent, state one aspect of the overall plan that should be reassessed. [1 mark]

Section B | Image-based questions

Q48 An adult with diabetes describes burning pain in the shaded areas shown below. Examination confirms bilaterally reduced distal pinprick sensation in the same distribution. **[3 marks]**

Reported pain distribution - front view



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 8 | Original schematic; not to scale.

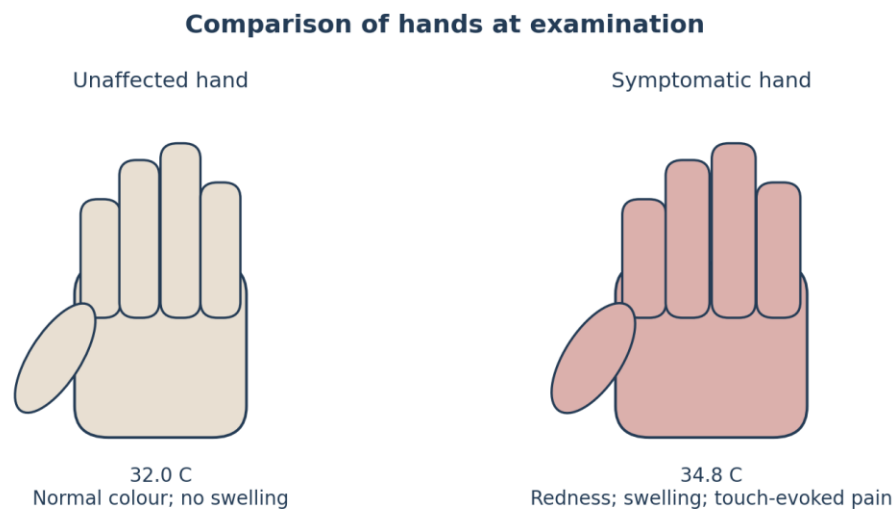
ffffff) Name the most likely neuropathic pain syndrome. [1 mark]

ggggggg) Describe the distribution using a standard clinical term. [1 mark]

hhhhhhh) Name one guideline-supported initial medicine for this condition, assuming no contraindication. [1 mark]

Section B | Image-based questions

Q49 Eight weeks after a wrist fracture, a patient reports disproportionate regional pain, touch sensitivity, colour/temperature changes, sweating change and reduced movement. Examination confirms the illustrated findings and restricted movement; no alternative diagnosis better explains them. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 9 | Original schematic; not to scale.

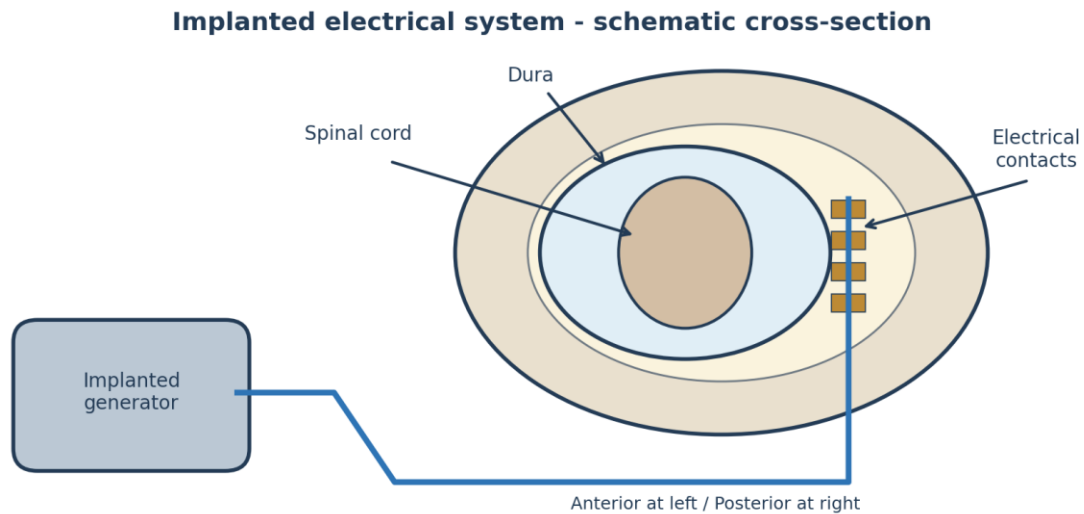
iiiiiii) Name the syndrome most strongly suggested. [1 mark]

jjjjjjjj) Which Budapest diagnostic category includes the illustrated temperature and colour asymmetry? [1 mark]

kkkkkkkk) State one core non-pharmacological component of management. [1 mark]

Section B | Image-based questions

Q50 This original schematic shows an implanted generator connected to electrical contacts behind the spinal cord. It is not a surgical-placement guide. **[3 marks]**



Original teaching schematic / fictional observations - not a patient image or procedural guide

Figure 10 | Original schematic; not to scale.

IIIIIIII) Identify the treatment modality. [1 mark]

mmmmmmmm) Name the anatomical compartment in which the illustrated spinal contacts lie. [1 mark]

nnnnnnnn) Name one important assessment domain before permanent implantation, other than the pain score. [1 mark]

Section C | Short-answer questions

Answer briefly. One correct term, example or concise statement is sufficient where requested.

Q51 State the usual duration threshold used to define chronic pain. **[1 mark]**

Q52 Name one material risk that should be discussed before a proposed spinal pain procedure. **[1 mark]**

Q53 Name the enzyme family inhibited by non-steroidal anti-inflammatory drugs. **[1 mark]**

Q54 Name one overdose-risk mitigation measure for a higher-risk patient prescribed an opioid. **[1 mark]**

Q55 Name one medication class that must be specifically reviewed for bleeding risk before a neuraxial pain procedure. **[1 mark]**

Section C | Short-answer questions

Answer briefly. One correct term, example or concise statement is sufficient where requested.

Q56 During a local anaesthetic injection, a patient develops circumoral numbness, tinnitus and a seizure. Name the suspected complication. **[1 mark]**

Q57 Name one non-analgesic-drug treatment used specifically to relieve pain from bone metastases. **[1 mark]**

Q58 Name one validated questionnaire used to measure disability associated with low back pain. **[1 mark]**

Q59 State one way activity pacing can be incorporated into a chronic pain rehabilitation plan. **[1 mark]**

Q60 Name one psychological domain that should be assessed before spinal cord stimulation. **[1 mark]**

END OF WRITTEN PAPER - 80 MARKS